KARTIK DHASMANA

Delhi -110096 kartikdhasmana@gmail.com + 91 9205306086 linkedin github/kartikdhasmana

Best Model Recommendation Report

Which model gave the best results?
Best Model: spaCy Pattern Matcher

Why spaCy Is the Best:

- Extracted the most valid results (91/200 JDs)
- Results were numerically reasonable (e.g., 2–5 years)
- No hallucinations or misinterpretations
- Fast and efficient — doesn't require large GPU or long inference times
- Highly customizable with pattern tweaks

Why Other Models Are Not Ideal:

MODEL 1: BERT QA

- Often guessed unrelated numbers
- Produced outliers (e.g., 40,000 years of experience)
- Sensitive to phrasing in the JD
- Needs significant post-processing to normalize output

MODEL 2: RoBERTa QA

- Similar issues to BERT:
  1. Huge outliers
  2. Extracted unrelated digits
- Output inconsistent across similar JDs
- Slower than spaCy

MODEL 3 : TARS (Zero-shot Classification)

- Clean and consistent output (always 1, 4, 8, or 12 years)
- But only worked on 42/200 JDs (low coverage)
- Limited to predefined ranges (not flexible for edge cases)

MODEL 4 : DistilBERT NER

- Focuses on named entities (like person names, orgs)
- Missed or ignored numeric patterns tied to experience
- Gave poor results unless experience phrases were very clearly worded

MODEL 5 : Prompt-based Heuristic

- Simple logic → fast
- But easily confused by context (e.g., unrelated numbers like salary, counts)
- Gave massively wrong outputs (up to billions!)
- No understanding of meaning — just a word search

Challenges Faced in Model Comparison:

- Absence of Ground Truth Labels
- Diversity in Job Description Formats
- Hallucination and Misinterpretation by QA Models
- Extreme Outliers in Output
- Difficulty in Normalizing Outputs
- Slow Performance of Transformer Models